UW–IBM Quantum Hackathon 2026

When Does Quantum Data Encoding Actually Help?

A Dimensionality Paradox in Projected Quantum Kernels

Track 0 · Track 2A · Track 2B · Track 2C

The Problem

Slot 1
(15 options)

Slot 2
(15 options)

Slot 3
(15 options)

Slot 4
(15 options)

↓

CLASSICAL

60-dim one-hot encoding

↓

RBF Kernel SVM

VS

QUANTUM

60 qubits → measure X, Y, Z

↓

180-dim projection

↓

RBF Kernel SVM

~50,000	246	172 / 74	Binary
possible designs	recipes tested	train / test split	kills cancer or not

Our Hypotheses

Based on the literature, we predicted:

H1

Quantum Outperforms Classical

Quantum encoding captures richer feature structure through entanglement, leading to higher classification accuracy.

Utro et al. (2025)

H2

Advantage Grows with Less Data

The geometric difference $g > \sqrt{N}$ signals quantum advantage — so with scarce training data, quantum should help more.

Huang et al. (2021)

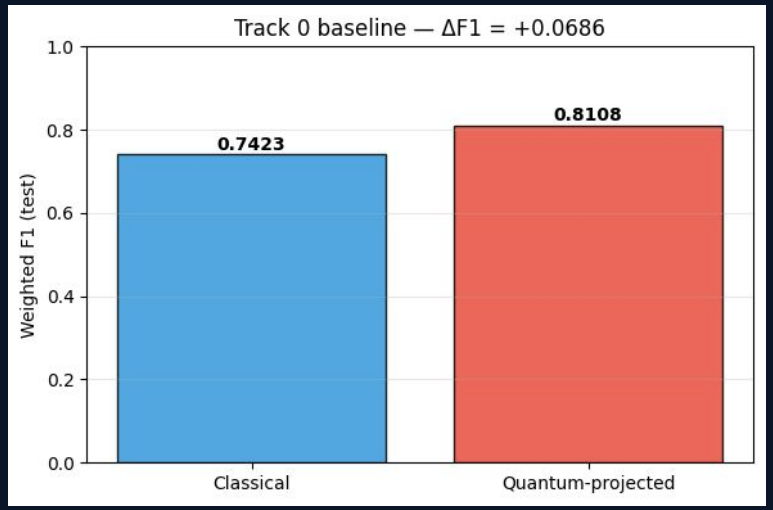
H3

Quantum Features Are Noise-Robust

The high-dimensional quantum projection should provide some resilience against label corruption and measurement noise.

General expectation

Track 0: Results



Test F1

+6.86%

Quantum WINS

0.8108 vs 0.7423

CV F1

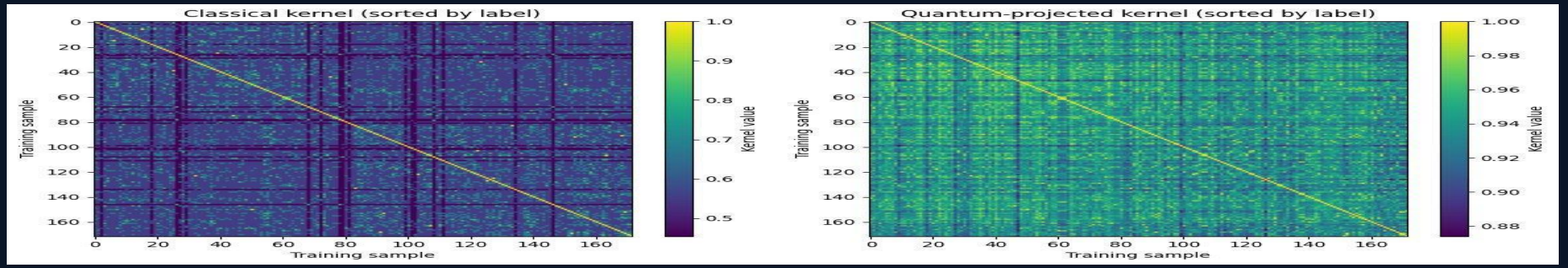
−6.3%

Classical WINS

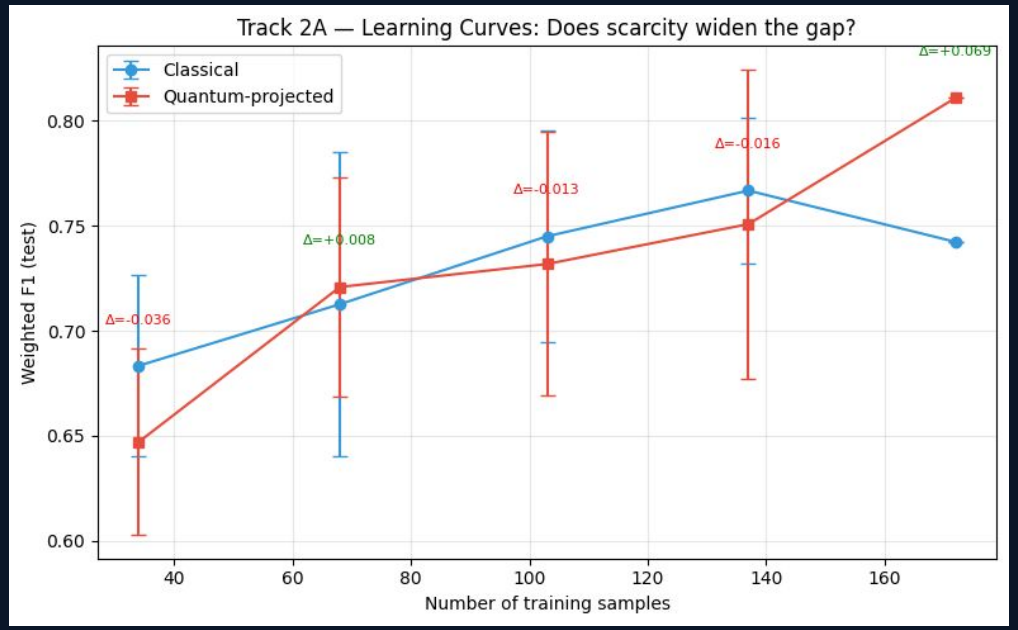
p < 0.0001 (permutation test)

Quantum generalizes better · Classical memorizes training patterns better

Kernel structure (sorted by label):



Track 2A: Learning Curves



ΔF1 = Quantum − Classical

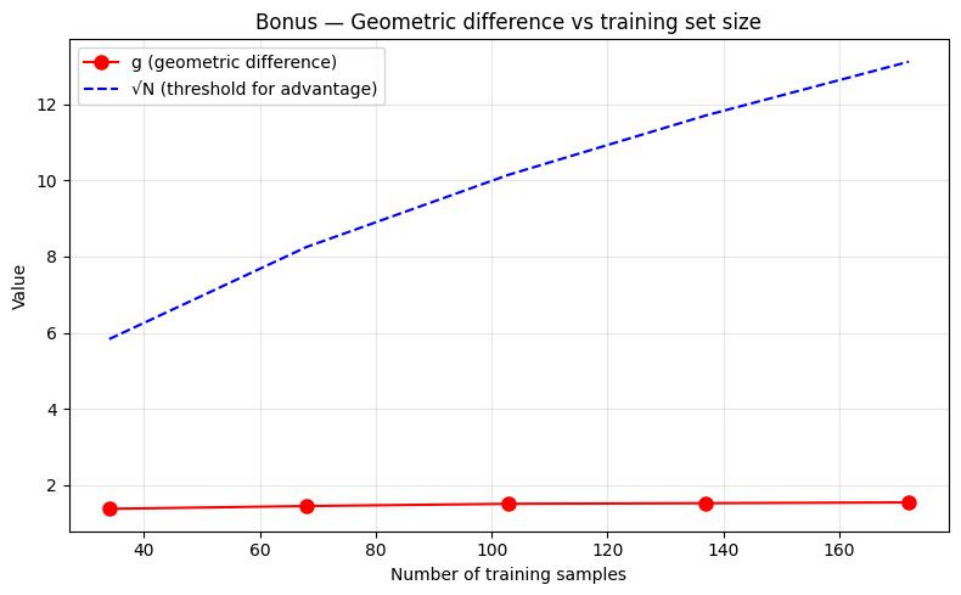
34 samples	−3.6%	Classical
68 samples	+0.8%	Quantum
103 samples	−1.3%	Classical
137 samples	−1.6%	Classical
172 samples	+6.9%	Quantum ✓

Hypothesis 2

REJECTED

Quantum only wins at full 172 samples.
180-dim features suffer from curse of dimensionality
with small training sets.

The Dimensionality Paradox



g / √N ratio by training size

N = 34

g/√N = 0.236

Theory: MORE potential

Reality: Quantum LOSES

N = 172

g/√N = 0.118

Theory: LESS potential

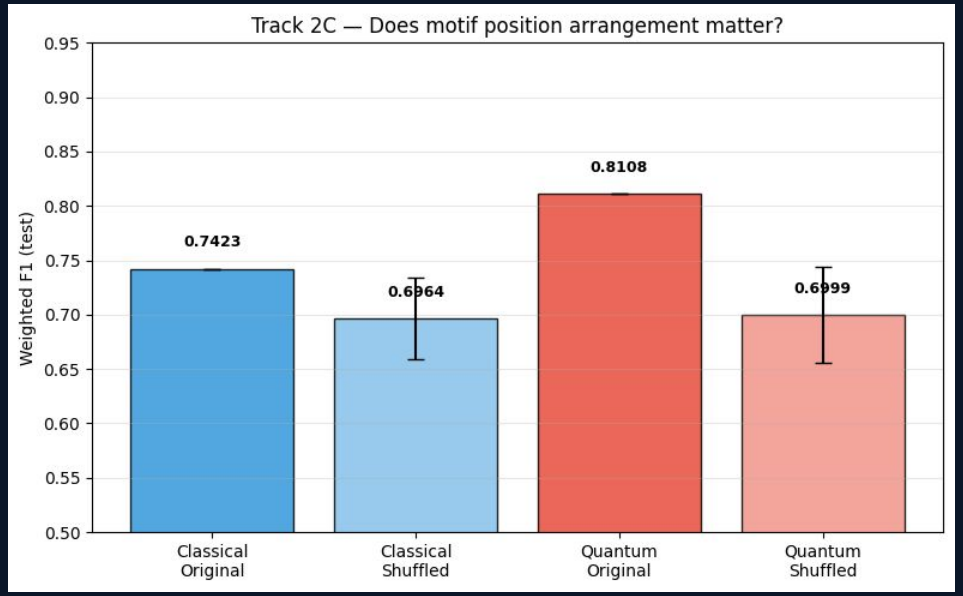
Reality: Quantum WINS

The same 180-dim projection that gives quantum its power is what kills it with only 34 samples.

Theory and practice contradict

Richer features demand more data to learn from

Track 2C: Position-Shuffling Ablation



After position shuffling:

Quantum drops

−11.1%

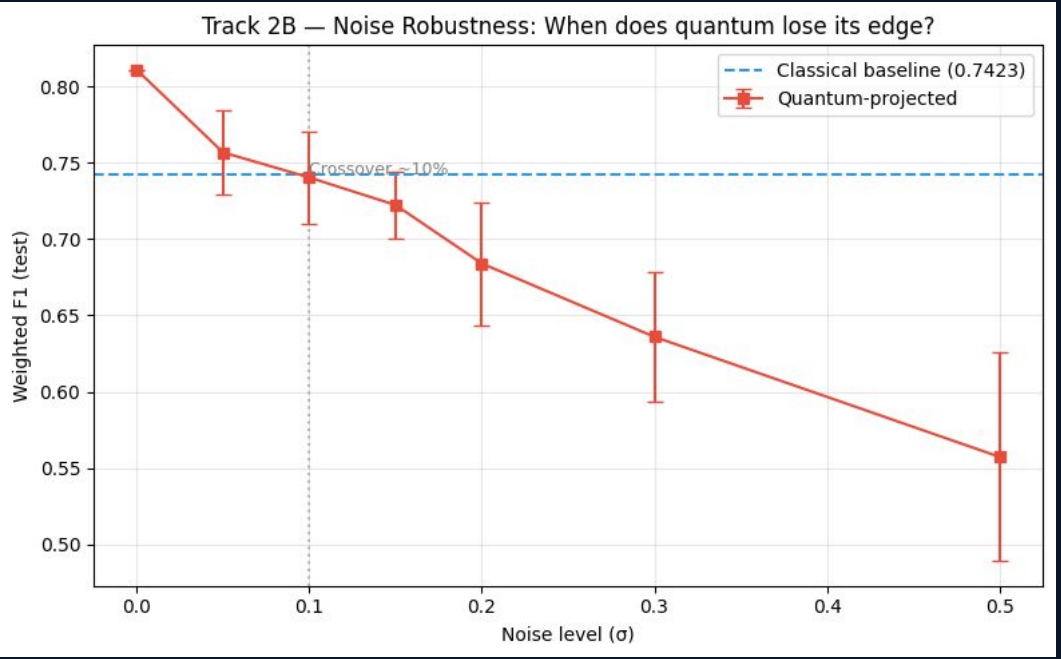
Classical drops

−4.6%

What this proves

Quantum's advantage comes specifically from cross-position entanglement. The ZZ feature map encodes how ingredients in different slots interact — destroy that structure and the advantage disappears entirely.

Track 2B: Noise Robustness



Advantage depends on precise quantum measurements.
At 10% noise: advantage gone. At 50%: quantum collapses far below classical.

Key Thresholds

0% noise

+6.86% advantage

10% noise

Advantage gone

50% noise

Q drops to 0.557

Hypothesis 3

REJECTED

Conclusion

Quantum encoding improves CAR T-cell prediction by 7% through cross-position entanglement.

BUT this advantage requires:

1. **Sufficient training data**

   >80% of dataset required — contradicts sparse-regime theory

2. **Precise measurements**

   <10% noise tolerance — advantage vanishes above this threshold

3. **Preserved position structure**

   Entanglement captures cross-slot interactions — shuffling destroys the advantage

The Dimensionality Paradox: PQK's richer features demand more data, challenging the assumption that quantum helps most when data is scarce.